# When does an LLM trust a specialist model? A cost-aware trust-routing audit — and why "calibrated reliability interfaces" are not a free win

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## Abstract

Frontier AI-for-science systems increasingly place a general LLM *above* specialist scientific models (protein, genome, single-cell), where the LLM must decide — under a verification cost — whether to **trust** a specialist's output, **verify** it, fall back to a **cheap baseline**, or **defer**. We audit that trust-routing decision and ask whether converting a specialist's reliability evidence into an explicit *calibrated reliability interface* improves routing over simply showing the raw confidence.

Across the initial three substrates (a GEARS/Norman perturbation dry-run, a single-cell foundation-model cue, and protein-structure prediction with Boltz-2/pLDDT) the robust finding is **cautionary, not celebratory**. (1) The LLM reasoning layer is strongly **cue-sensitive**: shown any reliability signal it routes far better than with none, but it also follows misleading signals. (2) On a *validated, calibrated* signal (pLDDT, which predicts true lDDT at Pearson 0.89 for monomers), an LLM (Claude Sonnet 4.6) converts it into much better cost-aware routing — **but the benefit comes from the calibrated *information*, not a directive recommendation**, and **packaging it as a "reliability card" does not robustly beat showing the raw confidence number** (Δnet +0.025, 95% CI [−0.037, +0.087]). (3) The card is **model-dependent and can backfire**: Claude Opus 4.8, a more risk-averse model, **over-verifies** under the card (97.5% verify rate) and is *significantly worse* than with raw confidence (Δnet −0.225, CI [−0.325, −0.113]), replicating the "stronger reasoning != better cost-aware orchestration" pattern. We conclude that **raw calibrated confidence is the robust lever; prompt-visible reliability *interfaces* are not a free win and are model-dependent** — presentation alone is insufficient, motivating enforcement-based (tool/MCP/post-training) routing.

A follow-up **variant-effect DMS** arm tests the same claim on the first substrate that passes the full trust-routing precondition: the specialist confidence signal is validated against experimental truth, the decision is non-saturated, and the signal adds modest information beyond a cheap conservation baseline. The pre-registered primary result remains a null: a calibrated reliability interface does **not** robustly improve net utility over raw specialist score across three models and three verification costs. This makes the negative result more informative, not less: even when the signal is real and decision-relevant, prompt-visible presentation is insufficient.

Acting on that conclusion, we then **enforce** the routing decision instead of presenting it (Phase 4). A deterministic **calibrated-risk gate** beats free-form LLM routing and is **manipulation-robust**, and this **replicates on a fresh, strictly leakage-controlled substrate built end-to-end**

**for the test** — even though, under strict leakage, the specialist's interface confidence is only moderately predictive (ipTM→DockQ Pearson **0.41**, not the exploratory 0.84) and the calibration is transferred. Gate - free-form LLM is significantly positive and grows with the verification price ($\approx +0.07$ at $\lambda = 0.5$ to $\approx +0.20$ at $\lambda = 0.8$); the LLM is the weak link in every regime (it **over-verifies**, and it follows a *corrupted* reliability card even with the raw signal present to contradict it); and a **conformal-abstention** gate meets its distribution-free guarantee (realised false-accept $0.107 <$ alpha $= 0.20$). The one honest boundary the weak signal exposes: calibrated verification beats *naive trust-all* only when the signal is sharp enough for the price (it does at $\lambda = 0.5$, not at $\lambda = 0.8$), whereas the gate beats *free-form LLM routing* unconditionally. **Enforcement — not presentation — is the robust, manipulation-resistant trust layer; the routing decision should not live in free-form LLM text.**

## 1. Question

> What does an LLM reasoning layer trust when it reads a specialist scientific model's output, and can that trust be made calibrated, auditable, and useful under verification cost?

Action set per item: `trust_sfm | verify_assay | default_baseline | defer`. Reward: `net = correct - lambda*assays` ($\lambda$ = verification price; primary $\lambda = 0.5$).

## 2. What each phase found

- **Phase 0 (GEARS/Norman dry run, 642 non-leakage Sonnet requests).** Routing is strongly cue-sensitive and only weakly calibrated; additive-disagreement evidence helps, misleading "reliability cards" hurt, and self-reported rationales frequently cite unavailable evidence.
- **Phase 1 (single-cell scFoundation cue).** NO-GO. The scFoundation-derived edge cue predicts GEARS wrongness at AUROC 0.599 vs 0.576 for a random same-gene control — near-noise. A specialist-metric check found GEARS ≈ a cheap additive baseline (replicating Ahlmann-Eltze et al. 2025). You cannot study calibrated trust where the signal carries no validated information.
- **Phase 2 (protein structure, Boltz-2 / pLDDT).** A substrate where the specialist is excellent *and* emits a validated calibrated confidence — the focus of this report.
- **Follow-up: variant-effect DMS.** The first substrate in this audit to pass the full precondition gate (validated confidence, non-saturated decision, and added value beyond a cheap conservation baseline). It strengthens the same headline: raw confidence is the robust lever; a calibrated reliability interface is not a free win over raw score.

## 3. Phase 2 methods (brief)

- **Substrate.** 80 RCSB targets released 2026-06-17 (after Boltz-2's 2023-06 training cutoff → leakage-safe), 40 monomer + 40 complex; sequences + reference structures from RCSB; structures predicted with Boltz-2 on an A100/A40 (Cayuga).
- **Truth.** Superposition-free CA-lDDT of prediction vs experimental reference (Mariani et al. 2013), monomer single-chain and complex all-chain (interface-inclusive). *Caveat: home-grown CA-lDDT, not yet validated against OpenStructure; complexes use lDDT, not DockQ.*
- **Calibrated signal.** pLDDT→P(wrong) via leave-one-out isotonic regression. pLDDT predicts lDDT at **Pearson 0.89 (monomers), 0.16 (complexes)** — the designed

monomer→complex calibration gap, confirmed.

- **Offline gate.** A deterministic "verify iff calibrated-risk > $\lambda$" policy must beat trust-all + shuffled/inverted controls before any LLM call. *Caveat: it is degenerate at the standard lDDT >= 0.7 cutoff (Boltz-2 is correct on ~95% of recent targets) and was made to pass at lDDT >= 0.9 — a post-hoc choice; signal AUROC there is 0.89.*
- **LLM pilot.** A balanced **40-target subset (20 monomer + 20 complex)** of the 80-target benchmark, stratified across the lDDT range, × 5 arms = 200 requests, Sonnet 4.6 and Opus 4.8, scored against held-out lDDT-truth. (The full 80 targets are used for the offline calibration gate.) Arms: `no_signal`, `raw_plddt_shown`, `calibrated_risk_shown_no_recommendation` (information without a directive), `calibrated_interface_sl` (risk + recommended action), `inverted_reliability_interface_control`.

## 4. Results

### 4.1 Sonnet ($\lambda$ = 0.5, lDDT >= 0.9; target-bootstrap 95% CIs, seed 13)

| arm | net/target | Δ vs no_signal (95% CI) |
|---|---|---|
| calibrated_interface_shown | 0.738 | +0.487 [+0.338, +0.625] |
| calibrated_risk_shown_no_recommendation | 0.725 | +0.475 [+0.325, +0.613] |
| raw_plddt_shown | 0.713 | +0.463 [+0.300, +0.613] |
| inverted_reliability_interface_control | 0.388 | +0.138 [+0.025, +0.250] |
| no_signal | 0.250 | — |

- **Any confidence cue robustly beats no-signal** (+0.46–0.49, CIs exclude 0). With no signal Claude defers/over-verifies ("can't assess reliability").
- **A4 — the benefit is informational, not directive.** calibrated_interface vs calibrated_risk_*no_recommendation*: Δ +0.013, CI [+0.000, +0.037] ≈ 0.
- **The calibrated card does NOT robustly beat raw pLDDT.** calibrated_interface vs raw_plddt: Δ +0.025, CI [−0.037, +0.087] — crosses zero.
- **Regime-appropriate caution.** Under the card, verify rate is 0.55 on complexes vs 0.20 on monomers — more checking exactly where pLDDT calibration degrades.
- **Cue-sensitivity persists.** The inverted card degrades routing 0.738 → 0.388.

### 4.2 Cross-model (Opus 4.8, same 200 requests)

| arm | Sonnet net | Opus net | Sonnet verify | Opus verify |
|---|---|---|---|---|
| raw_plddt_shown | 0.713 | 0.713 | 0.225 | 0.275 |
| calibrated_interface_shown | 0.738 | **0.487** | 0.375 | **0.975** |
| no_signal | 0.250 | 0.150 | 0.500 | 0.300 |

Opus calibrated_interface vs raw_plddt: **Δ** −0.225, **CI** [−0.325, −0.113] — **robustly worse**. The risk-framed card triggers blanket over-verification in the more risk-averse model (it verifies even confident monomers). A4 holds for Opus too (card vs no-recommendation = 0).

### 4.3 Robustness

The Sonnet/Opus conclusions are **stable across lDDT cutoffs 0.5–0.9** (`results/phase2_preflight/cutoff_ro` calibrated_interface - raw is $\approx 0$ for Sonnet and $\approx -0.25$ for Opus at every cutoff — so the post-hoc 0.9 choice affects only the offline-gate variance, not the pilot conclusions. Across $\lambda$ = 0.2/0.5/0.8 the lower arms keep their order; the only change is that calibrated_interface and raw_plddt swap at the top under high cost (card-raw: +0.070 / +0.025 / −0.020 at $\lambda$ = 0.2/0.5/0.8) — which further supports that the card does not robustly beat raw confidence.

## 5. Interpretation

The robust, cross-phase, cross-model claim:

> The LLM reasoning layer is **cue-sensitive**; surfacing **raw calibrated confidence** is the robust lever for better cost-aware routing; repackaging it as a calibrated **reliability *interface*** is **not a free win and is model-dependent** — neutral for Sonnet and harmful for the more risk-averse Opus, which over-verifies. The benefit, where present, is **informational, not directive**. Presentation alone is insufficient; **enforcement** (tools, MCP constraints, or post-training) is the natural next lever.

This extends Turpin et al. (2023) — LLMs follow cues regardless of validity — into a constructive, *calibrated-signal* regime: a correctly calibrated cue helps (via its information), a misleading one hurts, and a "reliability card" framing can over-trigger a risk-averse model.

## 6. Relation to prior work

- **Proto / EvoDesign (Merchant, Guo, Viggiano, ... Hie; bioRxiv 2026.06.22.733870) — the most directly relevant system.** Proto is a high-level programming language for generative biology: an LLM/agent composes ~120 specialist models (Evo2, ESM3, AlphaFold3, Boltz-2, AlphaGenome, ProteinMPNN, …) into multi-objective design programs, treating each specialist's confidence as an energy to minimize (pi(x) ~ p(x)*exp(-f(x)/T)). It is the strongest existing validation of the hybrid "reasoning layer above specialist models" stack this project assumes – but it trusts specialist confidence UNCONDITIONALLY: no calibrated trust/verify/defer decision, no verification price, no audit of the reasoning layer's trust. Proto itself reports the gap we study – "structure-prediction metrics are plausibility filters, not guarantees of function"; "no single in-silico metric separated functional from non-functional designs" – and routes around it with multi-objective consensus plus tens of wet-lab tests. Two of its constructions are concrete,* uncalibrated* precedents for our actions: (i) its 8-stage CRISPR filter cascade runs cheap filters first and expensive AlphaFold3 last with short-circuiting – a hand-built, cost-ordered verify-vs-defer pipeline (our net = correct - lambda*assays objective, minus the calibration); (ii) its multi-oracle structure consensus (require Boltz-2 AND AF2 AND AF3 to agree) is an uncalibrated verify-by-consensus. Our contribution is the calibrated, cost-aware, pre-registered trust layer that Proto-class agentic systems currently lack.
- **Cue-following / unfaithfulness:** Turpin et al. 2023. We add a cost-aware routing game and a *validated-calibrated* cue.
- **Cost-aware routing / selective prediction:** ASPEST (Kim et al. 2023), Adaptive-RAG, FrugalGPT/RouteLLM, learning-to-defer (Madras 2018; Mozannar & Sontag 2020). We route over a *fallible scientific specialist* under an assay cost.

- **Specialist soundness:** Ahlmann-Eltze, Huber & Anders 2025 (perturbation FMs ≈ linear baselines — why single-cell failed here); Jumper et al. 2021 and the McGuffin-group calibration audit 2024 (pLDDT calibration — why protein structure worked); Cheng et al. 2023 (AlphaMissense — a runner-up substrate).

## 7. Limitations

(a) Offline-gate PASS engineered post-hoc (lDDT >= 0.9 + in-distribution LOO calibration); (b) home-grown CA-lDDT, unvalidated vs OpenStructure, complexes not DockQ; (c) low-stakes substrate (~95% correct at lDDT >= 0.7); (d) the model-visible packet contains only confidence (+ regime), so "uses the signal" is partly forced; (e) n = 40, single primary $\lambda$, single MSA source, one run per model, default-False template baseline. A pre-registered confirmatory redo addressing all of these is specified in `PHASE2_PREREGISTRATION.md`.

## 7b. Post-publication validation (2026-06-23, on-cluster)

Three follow-up checks on Cayuga directly test the limitations above against gold-standard tools (artifacts under `results/phase2_{ost,dockq,leakage}_*`):

- **Monomer truth — confirmed.** OpenStructure all-atom lDDT vs the home-grown CA-lDDT on the 40 monomers: Pearson **0.99**; pLDDT→OST-lDDT = **0.894**, confirming the headline 0.89 monomer calibration against the gold standard.
- **Complex truth — corrected.** The all-chain CA-lDDT is intra-chain-dominated, so the reported pLDDT→complex "calibration collapse" (0.16) was largely a **metric artifact**. Against gold-standard **DockQ** (40 complexes): pLDDT→DockQ = **0.44**, and the proper interface confidence **ipTM→DockQ = 0.77** — the specialist *is* well-calibrated on complexes; route on ipTM, not pLDDT. Re-scoring the v1 episodes with DockQ truth leaves the headline intact (interface-raw ≈ 0 for Sonnet, ≈ −0.25 for Opus).
- **Leakage — the "leakage-safe" claim does not hold.** An MMseqs2 search of all 80 targets vs the full PDB, date-filtered via RCSB, finds that ~**70% (56/80 at a 2024 cutoff) have a >=90%-identical pre-cutoff homolog**. Release-date curation did not prevent training leakage; the ~95% base rate is partly memorization. The pre-registered redo's homolog dedup is therefore empirically required, and the substrate must be re-curated for genuinely low-homology folds.

Net: the routing **headline survives**, the monomer metric holds, the complex-calibration story is corrected, and the substrate's leakage is the principal reason to execute the confirmatory redo rather than lean on v1.

**Redo update (executed).** A genuinely leakage-controlled set was then curated (MMseqs2 vs full PDB + RCSB dates: **92% of recent post-cutoff depositions carry a >=30% pre-cutoff homolog**; 21 non-redundant low-homology targets survive) and predicted with Boltz-2. It is **100%/100% correct at the pre-registered cutoffs** (monomer lDDT >= 0.7, complex DockQ >= 0.23). So Boltz-2 succeeds on recent novel folds too: the **low-stakes problem is intrinsic to recent high-resolution structures, not an artifact of leakage** (the redo separates the two confounds). By the project's own NO-GO discipline a full *confirmatory* pilot cannot run on a 100%-base-rate substrate — a real routing substrate needs specialist *failure* cases (harder modalities), motivating the enforcement-layer pivot. A bounded 3-model demo (Sonnet 4.6, Opus 4.8, GPT-4.1) makes the consequence concrete and is identical across all three: `calibrated_interface - raw_plddt` net = −0.476 on the saturated substrate (the risk card triggers blanket over-verification where trust-all is free) vs **+0.05** only once stakes are manufactured (lDDT >= 0.9) — fresh, leakage-controlled, cross-model evidence that **reinforces the headline** (raw confidence is the robust lever; the reliability interface is not a free win). See `results/phase2_redo_curation/`.

**Specialist-failure substrate (a substrate that can finally test the headline).** Leakage memorises a chain's *fold* but not its *docking*, so protein-**complex interface** prediction fails even when chains are leakage-clean. A leakage-controlled set of 14 hard complexes (DockQ truth, ipTM cue) finally carries stakes — **DockQ base rate 50% at >= 0.49, ipTM→DockQ 0.865** — unlike the saturated monomer set. An illustrative ipTM-cued 5-arm pilot (Sonnet 4.6, Opus 4.8, GPT-4.1; n = 14, 1 seed) gives point estimates *directionally consistent* with the headline — raw confidence is the best arm, the calibrated *interface* sits ≈ 0.07 below raw across all three models, and the inverted control degrades routing — **but every contrast's 95% bootstrap CI crosses zero, so nothing is statistically significant**, and the pilot uses in-distribution calibration (not the pre-registered held-out design). It is therefore *suggestive, not confirmatory*: it shows the substrate *can* test the question and that the lean is the project's way, while the actual claim awaits the pre-registered run (held-out calibration, >= 3 seeds, >= 6 arms, $\lambda$-sweep, larger n). See `results/phase2_failure_substrate/`.

**Pre-registered confirmatory pilot (executed).** That run was then done properly (n = 14, **held-out** isotonic calibration fit on the 40 v1 complexes, 6 arms incl. a competing-cue, 3 seeds, $\lambda$-sweep, Holm-Bonferroni; H2 = interface vs raw as the single primary; `results/phase2_confirmatory/`). It **confirms and sharpens the headline**: (1) **H2 is NULL at the primary $\lambda = 0.5$** for all three models — the calibrated reliability *interface* does not beat raw confidence even under held-out calibration (the registered prediction); (2) **A4 is exact** — `calibrated_interface` equals the no-recommendation arm ($\Delta = 0.000$, Holm-significant), so the benefit is the *information*, not the recommendation; (3) **H1 is significant** — raw - no_signal = **+0.34 (p = .001, Sonnet) / +0.35 (p = .005, GPT)** at $\lambda = 0.8$, surviving Holm correction: raw calibrated confidence *genuinely* helps when verification is costly (the project's first statistically significant routing result); (4) cue-following persists (inverted < calibrated, Holm-sig) and a misleading authority cue does not override the numbers (competing-cue ≈ raw). The interface effect is $\lambda$-dependent (helps slightly at low cost, hurts at high cost via over-verification). Presentation-layer interfaces add nothing over the raw calibrated number — **enforcement-based routing remains the motivated next lever.**

**Powered confirmatory (N = 57, the definitive run).** Re-running the same pre-registered design on the expanded leakage-controlled set (53% DockQ base rate; ipTM→DockQ 0.84; 0/2394 errors) makes the result **statistically established** across all three models: **H1 — raw confidence significantly helps over no_signal** at $\lambda >= 0.5$ (+0.13...+0.17, p <= .007 at $\lambda = 0.5$; +0.31...+0.37, p ≈ 0 at $\lambda = 0.8$; the project's first solidly significant routing result); **H2 — the calibrated reliability interface is significantly *worse* than raw** at meaningful cost ($\lambda = 0.5$: Sonnet −0.094 p = .029, Opus −0.123 p = .016; $\lambda = 0.8$: −0.23...−0.30, p ≈ 0 for all three), even under held-out calibration. So the headline holds with statistical force: **raw calibrated confidence is the robust lever; the presentation-layer reliability interface is not a free win and actively hurts at real verification cost** — motivating enforcement-based routing. See `results/phase2_confirmatory/`.

## 8. Variant-effect extension: the precondition-passing confirmation

The protein-structure arm established the routing headline, but it also exposed a larger methodological issue: calibrated trust-routing has stakes only if the specialist confidence signal is validated, the decision is non-saturated, and the LLM uses the signal in a calibrated way rather than merely following the prompt cue. A follow-up variant-effect arm was therefore run as a precondition-passing confirmation rather than as a search for a positive interface result.

### 8.1 Cross-substrate precondition view

| arm | specialist confidence | gate result | honest interpretation |
|---|---|---|---|
| Phase 0 GEARS/Norman dry-run | disagreement / reliability cards | LLM use failed: cue-sensitive | Reliability-card presentation can move the model in the wrong direction. |
| Phase 1 single-cell | edge cue $\rightarrow$ P(wrong) | failed validation | No validated confidence signal to route on. |
| Phase 2 protein structure | pLDDT / ipTM | validated, but often low-stakes | Confidence is real, but saturated decisions make presentation-layer gains fragile. |
| Unpublished docking triage | docking score $\rightarrow$ P(active) | failed usable-signal test | Score-to-probability signal added little decision-relevant information beyond base-rate structure. |
| Variant-effect DMS | zero-shot VEP score $\rightarrow$ P(call correct) | **passed G1-G3** | First substrate where the precondition holds; prompt-interface H2 is still null. |

### 8.2 Gate result

On ProteinGym v1.3 DMS substitution data (96 human assays; gene/assay-level resampling), zero-shot variant-effect predictors pass all three gates before any LLM routing result is interpreted:

- **G1 validated.** Per-assay AUROC vs DMS: EVE 0.767, TranceptEVE-L 0.770, ESM1v 0.770, ESM1b 0.749, GEMME 0.768; every CI lower bound is $> 0.70$.
- **G2 decision-relevant.** DMS base-rate median 0.50, IQR [0.50, 0.55], with 99% of assays in [0.10, 0.90].
- **G3 beyond cheap baseline.** VEPs beat Site_Independent conservation by +0.028 to +0.041 AUROC, with CIs excluding 0.

This gate pass should be stated with its caveat: Site_Independent is already a strong baseline (AUROC 0.713), so the headroom is modest. The claim is not that variant-effect confidence is overwhelmingly superior; it is that this substrate finally makes the trust-routing question meaningful under the project's own precondition discipline.

### 8.3 LLM routing result

The pre-registered primary contrast was H2: whether a calibrated reliability interface improves net utility over simply showing the raw score. It does not robustly do so. All nine model-by-cost confidence intervals cross zero:

| model | $\lambda$=0.2 | $\lambda$=0.5 | $\lambda$=0.8 |
|---|---|---|---|
| Haiku 4.5 | +0.075 [−0.025,+0.183] | +0.031 [−0.010,+0.073] | +0.017 [0.000,+0.050] |
| Sonnet 4.6 | +0.033 [−0.054,+0.133] | −0.010 [−0.104,+0.083] | +0.125 [−0.008,+0.263] |
| Opus 4.8 | +0.045 [−0.041,+0.141] | +0.031 [−0.115,+0.177] | +0.012 [−0.229,+0.242] |

The secondary contrasts sharpen the interpretation. The benefit is informational, not directive: the full calibrated card performs about the same as calibrated risk without a recommendation. Cue-sensitivity also persists: an inverted reliability card drives over-verification (at $\lambda = 0.5$: Sonnet 98%, Opus 82%, Haiku 42%, vs 0-21% under the correct calibrated card). The public conclusion is therefore the combination, not either piece alone: **the first precondition-passing substrate still yields a prompt-interface null, while adversarial cue sensitivity remains.**

### 8.4 What this does and does not claim

This extension strengthens the cross-phase thesis: **raw confidence is the robust presentation-layer lever; better reliability cards are not enough.** It does **not** claim that AlphaMissense clinical-VUS replication has been run (AlphaMissense belongs to ProteinGym's clinical benchmark, not the DMS benchmark used here). It also does **not** claim that enforcement-layer routing works yet. The variant-effect result motivates that next systems test: make calibrated routing binding through tools, MCP constraints, or post-training, then evaluate against inverted and competing-cue controls at matched assay cost.

## 9. Reproducibility

All code, tests (163 passing), and compact result artifacts are in the repo under `experiments/trust_cue_attribut`
Key artifacts: `results/phase2_preflight/{boltz_contract,boltz_smoke_report,calibration_gate,`
`calibrated_gate,interface_pilot_score,interface_pilot_robustness, interface_pilot_score_opus,i`
Pipeline: `phase2_curate_pdb.py` → Boltz predict → `phase2_truth.py` (CA-lDDT) → `phase2_records.py` → `phase2_calibrated_gate.py` → `phase2_interface_pilot.py` → `phase2_score_episodes.py` / `phase2_robustness.py`. Large JSONL (requests, episodes, structures) stay ignored under `hpc_outputs/`.

## 9. Phase 4 — enforcement (the headline's consequence, tested)

If presentation cannot produce calibrated, manipulation-robust routing (§4–§7b), the decision should be **enforced** rather than delegated to free-form LLM judgment. On the N = 57 hard-complex substrate (held-out calibration, 3 models, 3 seeds, $\lambda$-sweep, bootstrap CIs; `results/phase4_enforcement/`):

- **A deterministic calibrated gate significantly beats the free-form LLM** (gate - LLM = +0.10…+0.19 at $\lambda = 0.5$, +0.24…+0.37 at $\lambda = 0.8$, all CIs exclude 0): delegating the decision costs −0.10…−0.37.

- **The gate is manipulation-robust; the LLM is not** — under a misleading reliability cue the LLM's net drops +0.06…+0.24 (significant) while the gate (computing from the raw signal) drops 0.
- **The gate is robust out-of-distribution** — on antibody-antigen (ipTM→DockQ 0.84$→$0.61) the general-fit gate stays within ~0.03 of the in-regime oracle and ties/beats the LLM.
- **Even under adversarial miscalibration the LLM does not recover** — shown a *corrupted* reliability card alongside the truthful raw ipTM, the LLM **follows the corrupted card** and loses to the (also-corrupted) blind gate (−0.09…−0.24), i.e. cue-following persists even when disconfirming evidence is in hand — sharpening Turpin et al. (2023).

Across every regime tested, free-form LLM routing never beats enforcement, and no "defer-to-reasoning" boundary materialised. **The robust, manipulation-resistant trust layer is a deterministic calibrated gate; the routing decision should not live in free-form LLM text.** This positions the work against the security-policy enforcement of Progent (arXiv:2504.11703) / AgentSpec (ICSE 2026) — *calibrated-risk* enforcement over a fallible *scientific specialist* with manipulation-robustness as the explicit goal, a cell those security systems and the cost-aware cascade *Trust or Escalate* (ICLR 2025) leave open. This Phase-4a evidence is exploratory; the powered confirmatory — >=120 complexes, the three enforcement mechanisms (calibrated gate / constrained decoding / conformal abstention) compared head-to-head, a real `default_baseline`, and a pre-specified adversarial-corruption regime — is fixed in advance in `experiments/trust_cue_attribution/PHASE4_PREREGISTRATION.md`.

### 9b. Confirmatory on a fresh strict-leakage substrate — the enforcement findings replicate

The confirmatory was run on a substrate built end-to-end for the test and **disjoint** from the 57 exploratory complexes: 2000 post-cutoff hetero-complexes harvested from RCSB → an MMseqs2-vs-full-PDB + release-date leakage filter (no pre-cutoff homolog >= 30 % id / 0.8 cov) keeping **161/2000 (8.1 %) → 75** Boltz-2-predicted (`--use_msa_server`) and DockQ-scored before the public ColabFold MSA server stalled (N is MSA-limited from the pre-registered >= 120; the run is resumable). Base rate is genuine (DockQ >= 0.49 in 47 %). Details: `results/phase4_confirmatory/`.

The exploratory **headline did not fully replicate, and that is the value of the confirmatory**: under *strict* leakage control the specialist's interface confidence is only moderately predictive of truth — ipTM→DockQ **+0.41**, not the exploratory **+0.84** (not a resolution artifact; the res <= 3.0 subset is also +0.41) — and the v1-fit calibration transfers to the new set only weakly (Pearson(calibrated risk, wrong) = +0.39). The confirmatory thus tests the harder question: does enforcement still help when the signal is *weak*?

It does. **H4.1** (enforced gate > free-form LLM) replicates: gate - free-form = **+0.07/+0.07/+0.09 at $\lambda$=0.5 (all p < .04)** and **+0.17/+0.17/+0.20 at $\lambda$=0.8 (all p $\approx$ 0)**, growing with $\lambda$ exactly as in 4a; the LLM remains the weak link (it over-verifies). **H4.2** (manipulation-robustness) and **H4.3** (constraint is a net no-op — the LLM over-verifies, not over-trusts) replicate; **conformal abstention** beats free-form *and* meets its guarantee (realised false-accept **0.107 < alpha = 0.20**); and under inverted/shift/noise corruption with the raw signal present the LLM **still** follows the corrupted card (no recovery at $\lambda$ >= 0.5). The one honest new boundary the weak signal exposes: at the *highest* price the calibrated gate no longer beats the **naive trust-all baseline**

($\lambda$=0.8: trust-all +0.467 > gate +0.392 > free-form +0.220), though it still beats the LLM; at $\lambda$=0.5 the gate (+0.580) beats both baselines. **Enforcement's advantage over free-form LLM routing is robust to a weak signal; its advantage over naive baselines is conditional on calibration sharpness relative to the verification price.**

## How to cite

See `CITATION.cff`. This report is intended for archival via Zenodo (GitHub release → DOI); the pre-registration is intended for OSF.